

Product Requirements Document (PRD)

Full-Stack Developer Portfolio Website

1. Product Overview

Product Name

Personal Full-Stack Developer Portfolio

Product Type

Responsive Full-Stack Web Application

Product Goal

Build a professional, scalable, high-performance portfolio platform that showcases:

- Developer profile
- Technical skills
- Education
- Work experience
- Projects
- Achievements
- Certifications
- Resume
- Contact functionality
- Admin management capabilities
- Analytics and integrations

The platform should function as both: 1. A professional online presence 2. A dynamic full-stack application demonstrating engineering capabilities

2. Business Objective

The portfolio should:

- Increase recruiter visibility
 - Showcase real-world development capabilities
 - Demonstrate frontend + backend expertise
 - Improve personal branding
 - Generate internship/job opportunities
 - Serve as a central hub for projects and achievements
 - Support future scalability
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3. Target Audience

Primary Users

- Recruiters
- Hiring managers
- Startup founders
- Technical interviewers
- Clients
- Hackathon organizers

Secondary Users

- Fellow developers
 - Open-source collaborators
 - Students
 - Networking connections
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4. Core Features

4.1 Authentication System

Features

- Secure admin login
- JWT authentication
- Session management
- Password encryption
- Forgot password functionality
- Protected admin routes

User Roles

Admin

Can:

- Edit portfolio content
- Add projects
- Manage skills
- View analytics
- Respond to contact messages

Visitor

Can:

- View portfolio
 - Download resume
 - Send messages
 - Explore projects
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4.2 Landing / Hero Section

Features

- Professional introduction
- Animated headline
- Dynamic typing effect
- Profile image/avatar
- Social media links
- Resume download button
- CTA buttons

Functionalities

- Smooth entrance animations
- Responsive layout
- Dynamic role switching
- Theme-aware styling

Content Example

- Name
 - Designation
 - Short bio
 - Tech stack highlights
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4.3 About Section

Features

- Professional summary
- Career objective
- Technical journey
- Developer philosophy
- Skills overview

Functionalities

- Editable via admin panel
 - Rich text support
 - Responsive formatting
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4.4 Education Section

Features

- Timeline-based education display
- Institution details
- Degree information
- GPA/percentage
- Academic achievements

Functionalities

- Dynamic CRUD operations
 - Sort by year
 - Expandable details
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4.5 Experience Section

Features

- Internship experiences
- Freelance projects
- Club/team experiences
- Leadership roles

Functionalities

- Timeline UI
 - Company logos
 - Experience duration calculations
 - Dynamic updates
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4.6 Skills Section

Categories

Frontend

- React.js
- Next.js
- HTML5
- CSS3
- Tailwind CSS
- JavaScript
- TypeScript

Backend

- Node.js
- Express.js
- REST APIs
- Authentication

Database

- MongoDB
- MySQL
- PostgreSQL
- Firebase

DevOps & Tools

- Git
- GitHub
- Docker
- Postman
- Vercel
- Netlify

AI/ML

- Python
- TensorFlow
- Scikit-learn
- OpenAI APIs

Functionalities

- Animated progress bars
- Skill filtering
- Category tabs
- Hover effects
- Dynamic management

4.7 Projects Section

Features

- Featured projects
- Project cards
- GitHub links
- Live demo links
- Tech stack badges
- Screenshots/videos
- Project filtering

Functionalities

- Dynamic project fetching
- Pagination
- Search functionality
- Category filtering
- Modal project view

Required Project Fields

- Project title
- Description
- Technologies used
- Features
- Challenges solved
- GitHub repository
- Live deployment URL
- Images
- Status

4.8 Certifications Section

Features

- Certificate display
- Verification links
- Organization logos
- Skill tags

Functionalities

- Dynamic uploads
- Sorting
- Expand preview

4.9 Achievements Section

Features

- Hackathons
- Awards
- Competitions
- Leadership achievements
- Event participation

Functionalities

- Timeline view
 - Achievement badges
 - Dynamic management
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4.10 Resume Section

Features

- Resume preview
- Download resume
- Multiple resume versions

Functionalities

- PDF viewer
 - Cloud storage integration
 - Auto-update resume
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4.11 Contact Section

Features

- Contact form
- Email integration
- Social links
- Location display
- Availability status

Form Fields

- Name

- Email
- Subject
- Message

Functionalities

- Email notifications
 - Form validation
 - Spam protection
 - Success/error handling
 - Database storage
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4.12 Admin Dashboard

Features

- Content management
- Analytics dashboard
- Visitor tracking
- Message management
- Resume management
- Project CRUD operations

Functionalities

- Secure admin access
 - Rich text editing
 - Real-time updates
 - Image uploads
 - Database management
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4.13 Analytics System

Features

- Visitor count
- Traffic analytics
- Device analytics
- Page performance metrics
- Resume download tracking
- Contact conversion tracking

Integrations

- Google Analytics
- Custom dashboard analytics

4.14 Theme System

Features

- Dark mode
- Light mode
- Dynamic theme switching
- Persistent theme preference

Functionalities

- Local storage persistence
 - Smooth transitions
 - Theme-aware components
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4.15 SEO Optimization

Requirements

- Meta tags
- Open Graph support
- Structured data
- Sitemap
- Robots.txt
- Dynamic metadata

Goals

- High Google ranking
 - Better recruiter discoverability
 - Fast indexing
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5. Technical Requirements

5.1 Frontend Stack

Framework

- React.js / Next.js

Styling

- Tailwind CSS
- Framer Motion
- CSS Modules

State Management

- Context API
- Redux Toolkit (optional)

Routing

- React Router / Next.js Router

UI Libraries

- Shadcn UI
 - Material UI
 - Lucide Icons
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5.2 Backend Stack

Runtime

- Node.js

Framework

- Express.js

API Style

- REST API

Authentication

- JWT
- bcrypt

File Uploads

- Multer
 - Cloudinary
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5.3 Database

Primary Database

- MongoDB

Optional Databases

- PostgreSQL
- Firebase

Collections/Tables

- Users
 - Projects
 - Skills
 - Experience
 - Education
 - Certifications
 - Contacts
 - Analytics
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5.4 Deployment

Frontend Hosting

- Vercel
- Netlify

Backend Hosting

- Render
- Railway
- AWS

Database Hosting

- MongoDB Atlas
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5.5 DevOps & CI/CD

Features

- GitHub Actions

- Automated deployment
 - Environment variable management
 - Version control
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6. Non-Functional Requirements

Performance

- Page load under 3 seconds
- Optimized images
- Lazy loading
- Code splitting

Scalability

- Modular architecture
- Reusable components
- API scalability

Security

- HTTPS
- Input validation
- Rate limiting
- XSS protection
- CORS protection

Accessibility

- WCAG compliance
- Keyboard navigation
- Screen reader support

Responsiveness

- Mobile-first design
 - Tablet optimization
 - Desktop optimization
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7. API Requirements

Authentication APIs

- POST /api/auth/login
- POST /api/auth/register

- POST /api/auth/forgot-password

Project APIs

- GET /api/projects
- POST /api/projects
- PUT /api/projects/:id
- DELETE /api/projects/:id

Skills APIs

- GET /api/skills
- POST /api/skills

Contact APIs

- POST /api/contact
- GET /api/contact

Analytics APIs

- GET /api/analytics
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8. Database Schema Overview

Users Collection

- id
- name
- email
- password
- role
- createdAt

Projects Collection

- title
- description
- technologies
- githubUrl
- liveUrl
- images
- category
- status

- createdAt

Contact Collection

- name
 - email
 - subject
 - message
 - status
 - createdAt
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9. UI/UX Requirements

Design Principles

- Minimalistic
- Modern
- Professional
- Interactive
- Clean typography

Animations

- Scroll animations
- Hover effects
- Page transitions
- Loading skeletons

User Experience Goals

- Smooth navigation
 - Fast interactions
 - Clear CTAs
 - Strong visual hierarchy
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10. Application Architecture

Frontend Architecture

- Components
- Pages
- Layouts
- Hooks
- Context
- Services

- Utilities

Backend Architecture

- Controllers
 - Routes
 - Middleware
 - Models
 - Services
 - Utilities
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11. Folder Structure

Frontend

```
src/  
├─ components/  
├─ pages/  
├─ layouts/  
├─ hooks/  
├─ context/  
├─ services/  
├─ assets/  
├─ utils/  
├─ styles/  
└─ App.jsx
```

Backend

```
server/  
├─ controllers/  
├─ routes/  
├─ middleware/  
├─ models/  
├─ config/  
├─ services/  
├─ uploads/  
└─ server.js
```

12. Third-Party Integrations

Required Integrations

- GitHub API
- LinkedIn links
- EmailJS / Nodemailer
- Google Analytics
- Cloudinary

Optional Integrations

- OpenAI APIs
 - Chatbot assistant
 - Firebase notifications
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13. Future Enhancements

Phase 2 Features

- AI chatbot assistant
- Blog system
- Admin CMS
- Project recommendation engine
- Real-time chat
- Visitor authentication
- Portfolio analytics AI insights

Phase 3 Features

- Multi-language support
 - Mobile app
 - Portfolio templates
 - Team collaboration
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14. Success Metrics

Technical KPIs

- Lighthouse score above 90
- API response under 300ms
- SEO score above 90

Business KPIs

- Increased recruiter responses
 - More portfolio traffic
 - Higher engagement time
 - Resume downloads
 - Contact conversions
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15. Risks & Mitigation

Risk	Mitigation
Slow performance	Optimize assets and APIs
Security vulnerabilities	Use secure authentication and validation
Downtime	Use reliable cloud hosting
Spam messages	CAPTCHA and validation
Scalability issues	Modular architecture

16. Development Roadmap

Phase 1 — Foundation

- Setup frontend
- Setup backend
- Database integration
- Authentication

Phase 2 — Core Features

- Hero section
- About section
- Skills section
- Projects section
- Contact system

Phase 3 — Advanced Features

- Admin dashboard
- Analytics
- Resume management
- SEO optimization

Phase 4 — Deployment

- Testing
 - Optimization
 - CI/CD setup
 - Production deployment
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17. Testing Requirements

Frontend Testing

- Component testing
- UI testing
- Responsiveness testing

Backend Testing

- API testing
- Authentication testing
- Database testing

Security Testing

- JWT validation
 - Rate limiting
 - Input sanitization
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18. Final Deliverables

Deliverables

- Fully responsive portfolio
 - Backend API system
 - Admin dashboard
 - Database integration
 - Deployment setup
 - Documentation
 - Resume integration
 - SEO optimization
 - Analytics dashboard
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19. Conclusion

This portfolio platform is designed not just as a static personal website but as a production-level full-stack engineering showcase demonstrating:

- Frontend engineering
- Backend architecture
- Database design
- API integration
- Authentication systems
- Deployment workflows
- UI/UX engineering
- Performance optimization
- Scalable application development

The final product should reflect industry-level engineering standards and provide a highly interactive, professional, and technically advanced user experience.